

WEEK 4

Q1. Train a Multiple Linear Regression model to predict like_count using view_count, commentCount, and dislike_count. Report R^2 and Adjusted R^2 . (2+2)

Calculate like count for:

view_count	dislike_count	commentCount
• 392658	• 0	• 189
• 454505	• 69	• 143
• 199512	• 23	• 93
• 9028403	• 308	• 7769

Q2. Now for Q1, split the dataset (80/20) and evaluate model performance using R^2 and MAE. (2)

Q3. Train models on different train-test splits (e.g., 60-40, 80-20, 90-10). Plot the training and testing error. Comment which model is underfitting and overfitting. (2+1)

Q4. Calculate sentiment scores of video_title. Check correlation with like_count. Visualise. (1+2)

Q5. Manually implement Gradient Descent to minimize cost for a simple regression problem: predicting like_count from view_count. Plot the cost function's convergence. (2+2)

Q6. Standardize all numerical features before regression. Does it improve model performance in Q1? (1+1)

Q7. Make a new feature: title_length. Check correlation between title_length and like_count. (2)

Q8. Create a binary column is_hit:

- 1 if like_count > median
- 0 otherwise

Train a Logistic Regression classifier using view count and comment count to predict is_hit. Report accuracy, precision, recall, F1. (3)

Q9. Predict view counts based on timestamp for test_standup using original dataset to train the model. (2)

Q10. Use TF-IDF vectorization on video_title and use it in a linear regression model to predict view_count, for the data in file test_standup. (3)